

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTECPC303 - PROGRAMMING, DATA STRUCTURE AND ALGORITHM USING C

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is a non-linear data structure?
A) Stack B) Queue C) Linked List D) Tree
- What is the process of visiting each element in a data structure exactly once called?
A) Searching B) Sorting C) Traversing D) Inserting
- What differentiates a circular linked list from a singly linked list?
A) It has more nodes B) Last node points to the first node C) It is used only in sorting D) First node points to last node
- 4 In a stack implemented using arrays, which operation checks if the stack is full?
A) Push B) StackFull C) Top D) Pop
- 5 Which of the following expression notations does NOT require parentheses?
A) Infix B) Prefix C) Postfix D) Both B and C
- 6 Which data structure is most suitable for reversing a list?
A) Queue B) Tree C) Stack D) Hash Table
- 7 Which of these queue types solves the problem of unused space in a linear queue?
A) Double-ended queue B) Priority queue C) Circular queue D) Stack
- 8 Which traversal method visits the left subtree, root, and then right subtree
A) Pre-order B) In-order C) Post-order D) Level-order
- What is the maximum number of children a binary tree node can have?
A) 1 B) 2 C) 3 D) Unlimited
- 10 Which of the following is used to represent graphs?
A) Adjacency Matrix B) Binary Search Tree C) Priority Queue D) Hash Table
- 11 What is the worst-case time complexity of linear search in an unsorted array of size n?
A) $O(1)$ B) $O(\log n)$ C) $O(n)$ D) $O(n^2)$
- 12 Which of the following sorting algorithms has the best average-case time complexity?
A) Selection Sort B) Insertion Sort C) Merge Sort D) Bubble Sort

Q2. Solve the following.

- A) Distinguish between linear and non-linear data structures and justify your opinion on 6 which type would be more suitable for a programmer.
- B) Define a linked list. Draw the structure for a singly linked list, a two-way linked list, and a 6 singly circular linked list.

Q3. Solve the following.

- A) Define and represent a Stack in a data structure. Explain the operations of a stack with 6 advantages and disadvantages.
- B) Convert each of the following Infix expressions into Prefix and Postfix expressions 6
 - a) $A+B/C-D * E$
 - b) $[(A+B)+C/D * E^F/G]$

Q4. Solve Any Two of the following.

- A) Write the C program for the following operations on a simple queue 6
 - a. Inserting an element
 - b. Deleting an element
- B) What is a queue? Explain three types of it with a suitable diagram. 6
- C) Consider the following queue, where the queue is a Circular Queue having 6 memory 6 cells.
Front = 2, Rear = 4. Queue: $_ , A , C , D , _ , _$.
Describe the queue as following operation take place : 1) F is added 2) Two letters are deleted 3) R is added to the queue 4) S is added to the queue 5) One letter is deleted.

Q5. Solve Any Two of the following.

- A) Write and explain any three important technologies used in the tree concept and the 6 graph concept in data structures, with examples.
- B) Explain the Binary Search Tree stepwise with the help of the elements given below 6
 $47, 12, 75, 88, 90, 73, 57, 1, 85, 50, 62$
- C) Explain any one type of graph traversal using an example. 6

Q6. Solve Any Two of the following.

- A) Write an algorithm for 6
 - a. Linear search
 - b. Binary search
- B) Write an algorithm and draw a flow diagram for the Bubble sort technique. 6
- C) Compare the mentioned asymptotic analysis Big-O, Big-Theta, and Big-Omega in 6 evaluating algorithm performance.

*** End ***